



Cairo University
Faculty of Engineering

EPES303-Electric Drive Systems

Lecture report (Traction system)

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Abstract

This report presents the design and analysis of an advanced traction system for a mid-size sedan. The focus is on optimizing the vehicle's performance by carefully selecting components based on specific parameters such as body weight, maximum velocity, acceleration, revolutions per minute (RPM), and power requirements.

The chosen mid-size sedan with a target velocity and acceleration. The powertrain design revolves around two potential motor choices: Permanent Magnet Synchronous Motors (PMSM) and Brushless DC Motors (BLDC). Each motor type is evaluated based on its efficiency, power density, and thermal performance. For energy storage, examining their energy density, lifecycle, and charging characteristics. Through simulations and comparative analysis, the study aims to determine the most suitable motor and battery combination to achieve optimal performance, efficiency, and reliability.

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Introduction

The rapid advancements in electric vehicle (EV) technology have highlighted the importance of efficient and powerful traction systems. A critical component of these systems is the electric motor, which directly influences the vehicle's performance, efficiency, and overall driving experience. This study aims to investigate the traction system of a mid-sized sedan, focusing on the calculation of rotational speed (RPM) and power requirements. To achieve this, we will delve into the principles of vehicle dynamic, we can accurately determine the power and speed characteristics necessary for optimal vehicle performance.

Through a comparative analysis of PMSM and BLDC motors, we aim to provide insights into their respective advantages and limitations. The selection of the appropriate motor technology will be based on a comprehensive assessment of performance, efficiency, cost, and ease of control. This investigation will not only contribute to the academic understanding of traction systems but also offer practical recommendations for the design and development of efficient electric vehicles.

In addition to the motor selection, the energy storage system is a critical component of the traction system. The analysis considers factors such as energy density, lifecycle, thermal management, and integration with the selected motor type.

Through a series of simulations and detailed comparative analyses, this research aims to identify the optimal combination of motor and battery that will deliver the desired performance for the mid-size sedan. The findings of this study are expected to provide valuable insights and contribute to the advancement of traction system designs for future electric vehicles.

System requirements

First, we need to define our system parameters.

Our design and calculations will be done based on the following specifications and assumptions:

- Size: Mid-size sedan car.
- Acceleration: 0 – 100 km/hr. in around 7 seconds.
- Range: 500 km.
- Maximum speed: 200 km/hr.
- Driving conditions: Combination of both city driving and highway driving.
- Weight of the car(without passengers): 1500kg
- Approximate total weight of the car: 1850 kg

*Hint: the approximate total weight is calculated based on fully loaded passengers and the car boot (trunk) is also fully loaded.

- This car will target most families (small and medium-sized families) and young people.
- It also will be suitable for daily use and offers a kind of sport performance.
- Tire size : 18" = 0.4572 m



Motor Selection

Description:

To choose the suitable motor that can achieve our requirements, we need to choose the suitable motor.

The two main parameters that we should consider while choosing an electric motor are the power and the torque, so, we need to calculate the needed power and torque.

Power Formulas

$$P = (F * v) / 1000$$

Where:

- P: Power in kW
- F: Force in Newton (N)
- v: Velocity in m/s

Where the Force (F) can be estimated from the drag force and rolling resistance.

$$\text{Force (F)} = F_{\text{drag}} + F_{\text{rr}}$$

Where:

- F_{drag} : Drag force
- F_{rr} : Rolling resistance force

The drag force is given by:

$$F_{\text{drag}} = 0.5 * \rho * A * C_d * v^2$$

Where:

- ρ : Air density (approximately 1.225 kg/m³ at sea level)
- A: Frontal area of the vehicle in m² , assumed to be 2.2 m²
- C_d : Drag coefficient (typically around 0.28 for a sedan)
- v: Velocity in m/s , based on maximum speed = 220 km/hr. = 55.6 m/s

The rolling resistance coefficient is given by:

$$F_{\text{rr}} = C_{\text{rr}} * m * g$$

Where:

- C_{rr} : Rolling resistance coefficient (typically around 0.01 to 0.02)
- m = Mass of the vehicle in kg
- g = Gravitational acceleration (approximately 9.81 m/s²)

Power calculations

$$F_{\text{drag}} = 0.5 * 1.225 * 2.2 * 0.28 * (55.56)^2 = 1164.7 \text{ N}$$

$$F_{\text{rr}} = 0.015 * 1850 * 9.81 = 272.2 \text{ N}$$

$$F = 1436.9275 \text{ N}$$

$$P = (1436.9275 * 55.56)/1000 = 79.8 \text{ kW}$$

Torque Formulas

$$T_{\text{acceleration}} = (m * a * r_w) / \eta$$

Where:

- $T_{\text{acceleration}}$: Acceleration torque in N.m
- m : Mass of the vehicle in kg
- a : Acceleration in m/s^2 (convert from 0-100 km/h to m/s^2)
- r_w : Radius of the wheel in meters
- η : Efficiency of the drive system (typically around 0.85 to 0.90)

Torque Calculations

$$T_{\text{acceleration}} = (1850 * a * 0.4572) / 0.9$$

$$a = (55.62 / (2-0)) / 6 = 4.63 \text{ m}^2 / \text{sec}.$$

$$T_{\text{acceleration}} = (1850 * 4.63 * 0.4572) / 0.9 = 4351.274 \text{ N.m}$$

*The Torque is high, so we need to put a gearbox to reduce the torque required by the motor.

GR: Gear ratio (ratio between driven gear and driving gear)

$$\text{GR} = 11:1$$

$$T = 395.57 \text{ N.m (required by the motor)}$$

Motor types

To match our requirements (torque, weight and power), we found that the best choices for our required motor are both Permanent Magnet Synchronous Motors (PMSM) and Brushless DC Motors (BLDC).

Permanent Magnet Synchronous Motors (PMSM)

Permanent Magnet Synchronous Motors (PMSM) are a popular choice in modern electric vehicle (EV) applications due to their high efficiency and performance.

Key Features of PMSM

1. High Efficiency

PMSMs are known for their high efficiency, often achieving over 90%. This efficiency is due to the use of permanent magnets in the rotor, which reduces energy losses.

2. Smooth Torque Delivery

These motors provide smooth and consistent torque with minimal ripple, which is ideal for applications requiring precise control and stable operation.

3. High Power Density

PMSMs have a high power-to-weight ratio, making them compact and lightweight. This is beneficial for EVs where space and weight are critical factors.

4. Reliable Performance

PMSMs offer reliable performance over a wide range of speeds. They maintain high torque even at low speeds, which is crucial for starting and accelerating.

5. Low Maintenance

- Due to the absence of brushes and commutators, PMSMs require less maintenance compared to brushed motors, leading to longer operational life and reduced downtime.

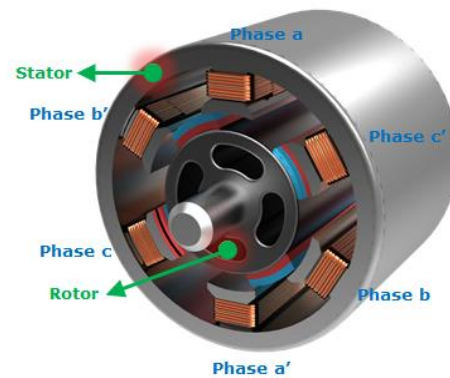
Control Systems

- Field-Oriented Control (FOC): PMSMs typically use advanced control techniques like Field-Oriented Control (FOC) or Vector Control to achieve precise and efficient operation.
- Sinusoidal Waveform: The control system uses sinusoidal waveforms to drive the motor, reducing torque ripple and improving overall performance.

Advantages & Disadvantages

Advantages

- High efficiency and power density
- Smooth and consistent torque
- Reliable and low maintenance



Disadvantages

- Higher cost compared to some other motor types due to complex control systems and the use of rare-earth magnets
- More complex control algorithms required

PMSMs are a robust choice for applications demanding high performance and efficiency, making them ideal for electric vehicles and other high-end applications.

Brushless DC Motors (BLDC)

Key Features of BLDC

1. High Efficiency

- BLDC motors are known for their high efficiency. They typically achieve efficiencies of 85-90% and can be very effective in converting electrical energy into mechanical energy with minimal losses.

2. High Reliability

- BLDC motors have fewer mechanical parts compared to brushed motors (no brushes or commutators). This results in less wear and tear, leading to increased reliability and longevity.

3. Excellent Speed Control

- BLDC motors offer precise speed control, which is beneficial for applications requiring variable speeds. They are often used with electronic controllers to achieve precise motion control.

4. High Torque-to-Weight Ratio

- BLDC motors provide a high torque-to-weight ratio, making them compact and suitable for applications where space and weight are critical factors.

5. Low Maintenance

- Since there are no brushes to replace, BLDC motors require less maintenance, which reduces downtime and maintenance costs over the life of the motor.

Control Systems

➤ Trapezoidal Waveform

BLDC motors typically use trapezoidal waveform control, which is simpler than the sinusoidal control used in PMSMs.

➤ Electronic Speed Controllers (ESC)

BLDC motors are often paired with ESCs for precise control of speed and torque.

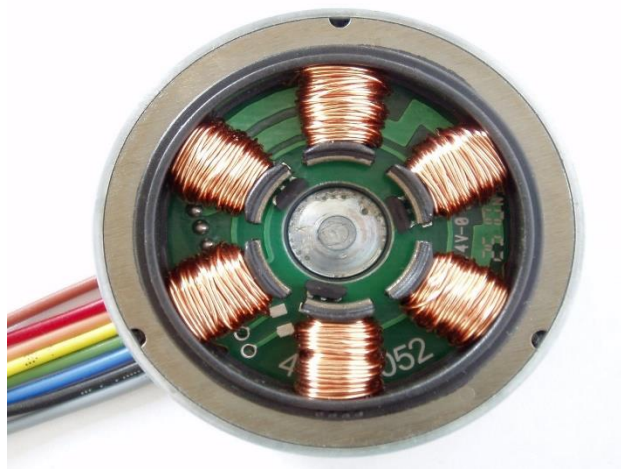
Advantages & Disadvantages

Advantages

- High efficiency and reliability
- Good speed control and high torque-to-weight ratio
- Low maintenance requirements

Disadvantages

- Slight torque ripple due to trapezoidal control
- Lower efficiency compared to PMSMs in some applications
- Potentially higher noise levels at low speeds



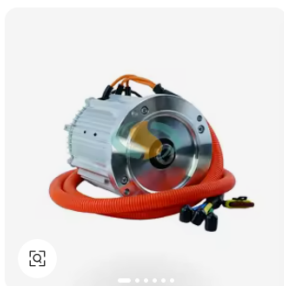
Permanent Magnet Synchronous Motors (PMSM) VS Brushless DC Motors (BLDC)

Both Permanent Magnet Synchronous Motors (PMSM) and Brushless DC Motors (BLDC) are excellent choices for electric vehicle (EV) applications, but they each have unique strengths. PMSMs are known for their high efficiency, smooth torque delivery, and high power density, making them ideal for high-performance applications. They utilize complex control systems like Field-Oriented Control, which allows for precise and stable operation, albeit at a higher cost and complexity. On the other hand, BLDC motors are highly efficient and reliable with a simpler control system based on trapezoidal waveforms. While they exhibit slight torque ripple and potentially higher noise at low speeds, they are often more cost-effective and easier to integrate, making them suitable for a wide range of applications including small EVs, robotics, and industrial automation.

Suitable Option for your EV

Given your requirements for a mid-sized EV with 80 kW power and 400 Nm torque, a PMSM would be more suitable due to its high efficiency and smooth torque delivery, which is crucial for both performance and driving comfort. The precision and reliability of PMSMs will ensure your EV performs exceptionally well across various driving conditions.

Permanent Magnet Synchronous Motors (PMSM)



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Battery types

The most common type of car batteries are the lead-acid batteries which are divided into

Flooded lead-acid (FLA)

Known as wet cell batteries, are a type of rechargeable battery that have been around since the 1850s.

These batteries consist of lead plates submerged in an electrolyte solution made of water and sulfuric acid. When the battery is charged or discharged, a chemical reaction occurs between the lead plates and the sulfuric acid, producing lead sulfate and releasing electrical energy.

During charging, hydrogen and oxygen gases are produced, which need to be vented to prevent pressure build-up. Therefore they need to be placed in a well vented area

Maintenance

- They require regular maintenance, including checking and topping up the electrolyte levels with distilled water.

Cost

- Despite their weight and lower energy density compared to other battery types, they are relatively inexpensive to produce and are widely used in automotive and backup power applications.

Applications

- Commonly used in car batteries, backup power systems, and renewable energy storage systems.

Advantages

- Low cost and high power-to-weight ratio
- Reliable and well-understood technology

Disadvantages

- Heavy and bulky
- Requires regular maintenance
- Can produce explosive gases during charging

Absorbent Glass Mat (AGM)

A type of lead-acid battery that offers several advantages over traditional flooded lead-acid batteries. It uses a special glass mat to absorb the electrolyte, which makes them spill-proof. The electrolyte is absorbed between the plates and held in place by the glass mat. A chemical reaction occurs between the lead plates and the sulfuric acid electrolyte, producing lead sulfate and releasing electrical energy.

Maintenance

- Unlike flooded lead-acid batteries, AGM batteries do not require regular maintenance. They are sealed, so there's no need to check or refill the electrolyte.

Efficiency

- They offer higher efficiency, faster charging, lower self-discharge rate and a longer lifespan compared to traditional flooded lead-acid batteries.

Shock resistance

- They are more resistant to vibration and shock, making them ideal for high-performance vehicles and applications with advanced features.

Applications

- Commonly used in modern vehicles with advanced electrical demands, high-performance vehicles, backup power systems, and renewable energy storage systems.

Advantages

- Maintenance-free
- Spill-proof (Sealed) and safer to handle
- Higher efficiency and faster charging
- Longer lifespan
- Resistant to vibration and shock

Disadvantages

- More expensive than traditional flooded lead-acid batteries
- Heavier compared to some other battery types

Enhanced Flooded Battery (EFB)

An advanced version of traditional flooded lead-acid batteries. They are designed to meet the higher power demands of modern vehicles. They are similar to standard flooded batteries but include enhancements like carbon additives in the lead plates and a specially designed grid structure. These improvements help in better electron transfer and more consistent energy flow. They use a liquid sulfuric acid electrolyte that reacts with lead plates to produce electrical energy. They are sealed and use a Polyfleece material to line the lead plates, offering additional protection and durability. They store more energy and handle frequent charge and offer better discharge cycles.

Maintenance

- While they are still wet cell batteries, they require less maintenance than traditional flooded batteries

Applications

- Ideal for vehicles with start-stop systems, regenerative braking, and other high-demand electrical features.

Advantages

- Enhanced durability and lifespan
- Higher heat tolerance
- Better performance in high-demand applications
- Improved charge acceptance and cyclic durability
- Suitable for start-stop systems and regenerative braking

Disadvantages

- Higher cost compared to standard flooded batteries
- Still require some maintenance, though less than traditional flooded batteries.

Lithium-Ion (Li-Ion)

A type of rechargeable battery that have become increasingly popular due to their high energy density, long lifespan, and lightweight design. They are widely used in various applications, from consumer electronics to electric vehicles. Li-ion batteries consist of an anode (typically made of graphite), a cathode (made of a lithium metal oxide), and an electrolyte that allows lithium ions to move between the anode and cathode. During charging, lithium ions move from the cathode to the anode through the electrolyte. During discharging, the ions move back to the cathode, releasing energy.

Safety

- Modern Li-ion batteries include safety mechanisms to prevent overcharging, overheating, and short-circuiting.

Maintenance

- These batteries are sealed and do not require maintenance, such as topping up with water.

Applications

- Commonly used in smartphones, laptops, tablets, power tools, electric vehicles, and renewable energy storage systems.

Advantages

- High energy density and lightweight
- Long life cycle
- Long lifespan and low self-discharge rate
- Maintenance-free
- Available in various shapes and sizes to suit different applications

Disadvantages

- Higher cost compared to other rechargeable batteries
- Sensitive to high temperatures and overcharging, requiring built-in safety features
- Degradation over time, with performance decreasing after several hundred charge cycles

Nickel-Metal Hydride (NiMH)

Used in some older hybrid vehicles, NiMH batteries offer decent energy density but are generally heavier than Li-ion.

Safety

- Non-toxic: NiMH batteries do not contain toxic cadmium, making them safer for the environment.
- Stable: They are generally stable under normal usage conditions, but like all batteries, they should be handled with care to avoid short-circuiting or overheating.

Maintenance

- Low Maintenance: NiMH batteries require minimal maintenance compared to other rechargeable batteries.
- Regular Charging: To maintain their performance, it's important to charge them regularly and avoid letting them fully discharge.

Applications

- Consumer Electronics: Widely used in devices like digital cameras, laptops, and portable gaming devices.
- Hybrid Vehicles: Commonly used in hybrid cars due to their high energy density and reliability.

Advantages

- High Energy Density: NiMH batteries can store more energy for the same size compared to NiCd batteries.
- Environmentally Friendly: They are more environmentally friendly than NiCd batteries as they do not contain toxic cadmium.
- Good Cycle Life: They can be recharged and discharged many times without significant degradation

Disadvantages

- Memory Effect: NiMH batteries can suffer from a "memory effect" if not fully discharged before recharging.
- Self-Discharge: They tend to lose charge over time when not in use, although low self-discharge variants are available

Ultra capacitors

These store energy in an electric field rather than a chemical reaction. They offer very fast charging and discharging but have lower energy density compared to batteries.

Safety

- Ultra capacitors are generally safe to use. They are sealed units and, when handled properly, pose minimal health hazards². However, misuse, such as overcharging or overheating, can lead to the release of internal components, which might be hazardous.

Maintenance

- One of the biggest advantages of ultra-capacitors is that they require very little maintenance. They have no memory effects, high cycle stability, and can operate over a wide range of temperatures³. This makes them ideal for applications where maintenance is difficult or costly.

Applications

- Electric Vehicles: Providing quick bursts of power for acceleration and regenerative braking.
- Renewable Energy Storage: Storing energy from sources like solar and wind.
- Industrial Equipment: Offering backup power and improving efficiency.
- Medical Devices: Used in defibrillators and pacemakers.

Advantages

- High Power Density: They can deliver and absorb energy quickly.
- Long Lifespan: Can endure hundreds of thousands of charge-discharge cycles.
- Temperature Resilience: Operate effectively across a wide temperature range.

Disadvantages

- Lower Energy Density: They store less energy compared to batteries.
- Cost: Can be more expensive than traditional batteries.
- Size: Larger physical size for the same amount of energy storage.

Solid-State Batteries

These are a promising future technology that could offer higher energy density, faster charging, and improved safety compared to current Li-ion batteries.

Safety

- Reduced Fire Risk: Solid electrolytes are non-flammable, significantly lowering the risk of fires and explosions compared to liquid electrolytes.
- Thermal Stability: They provide enhanced safety and stability under various operating conditions.

Maintenance

- Low Maintenance: They degrade slower than traditional batteries, meaning they need fewer replacements.
- Advanced Systems: High-tech battery management systems are necessary to ensure optimal performance.

Applications

- Electric Vehicles (EVs): Provide longer driving ranges and increased safety, making them a key component in the future of EVs.
- Renewable Energy Storage: Efficiently store energy generated by renewable sources like solar and wind.

Advantages

- High Energy Density: Store more energy in a smaller volume, leading to longer usage times for devices and vehicles.
- Long Lifespan: They exhibit better cycle stability and lower degradation rates over time.

Disadvantages:

- High Cost: Currently more expensive to produce compared to traditional batteries due to complex manufacturing processes and advanced materials.
- Material Stability Issues: Challenges with the stability of solid electrolytes, particularly with newer materials.

Battery selection

The choice of battery technology depends on various factors, including vehicle type, desired range, charging infrastructure, and cost.

The most common type of battery used in electric vehicles (EVs) today is the lithium-ion (Li-ion) battery.

Why we select lithium-ion battery?

High Energy Density: Li-ion batteries pack a lot of energy into a small space, enabling longer driving ranges.

Lightweight: Compared to other battery types, they are relatively lightweight, which improves vehicle efficiency.

Fast Charging: Some Li-ion chemistries support fast charging, reducing the time spent waiting for a full charge.

Electric car battery 80kwh 100kWh, 80kw 500v battery for electric car, lithium ion 500v
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Variations

Total options: 1 Nominal Voltage ; 1 Nominal C... [Select now](#)

Nominal Voltage(1): 537v

537v

Nominal Capacity(1): 150Ah

150Ah

Battery Type(1)

pack

Shipping

Shipping solutions for the selected quantity are currently unavailable

Conclusion

In this study, we have meticulously analyzed the design of an advanced traction system for a mid-size sedan, focusing on key performance metrics such as maximum velocity, acceleration, and overall efficiency. By comparing Permanent Magnet Synchronous Motors (PMSM) and Brushless DC Motors (BLDC) alongside lithium-ion and sulphuric acid batteries, we aimed to identify the optimal combination to achieve superior performance. Through comprehensive simulations and detailed evaluations, the findings indicate that the PMSM, when coupled with a lithium-ion battery, provides the best overall performance in terms of efficiency, power density, and reliability for the specified vehicle parameters.

The results of this research offer valuable insights into the development of efficient and robust traction systems for electric vehicles. The combination of PMSM and lithium-ion batteries not only meets the performance goals but also ensures sustainability and long-term viability. These findings contribute to the broader field of electric vehicle design, highlighting the importance of careful component selection to enhance vehicle performance and efficiency. Future work could explore additional motor and battery technologies, as well as the integration of renewable energy sources to further advance the capabilities of electric vehicle traction systems.

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